Interaction Day II: PHYS627/ECE523

1. At the end of last class, we observed the non-uniqueness of a density matrix's pure state decomposition. Let's begin developing the physics behind this aspect of the formalism.

   (a) Confirm that the following mixed ensembles share a density matrix ($\rho_1 = \rho_2$)

   $$\rho_1 = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| \quad , \tag{1}$$

   $$\rho_2 = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| \quad , \tag{2}$$

   where $|\pm\rangle = |\pm\rangle_x$ are x-basis eigenstates.

   (b) Show that both pure-state mixtures can be created from the two-qubit maximally entangled Bell state $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ by projectively measuring one qubit and recombining the resulting single-qubit states into a single ensemble, i.e.,

   $$\rho_{1,2} = \sum_k \left(\langle e_k| \otimes \mathbb{I}_2\right) \rho_{\Phi^+} \left(|e_k\rangle \otimes \mathbb{I}_2\right) \tag{3}$$

   for $\{|e_k\rangle\}$ spanning two different measurement bases. (note: we will later connect this perspective to modeling quantum noise.)

2. Imagine you are given two ensembles of single-qubit quantum states, one pure, $|\psi\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$, and one mixed, $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$.

   (a) Describe an experimental protocol for distinguishing between the two ensembles.

   (b) What is the minimum possible (best-case) number of measurements that would provide conclusive distinction?

3. Consider the following family of pure-state convex decompositions,

   $$\rho(\alpha) = \frac{1}{2}|\psi_1\rangle\langle\psi_1| + \frac{1}{2}|\psi_2\rangle\langle\psi_2| \quad , \tag{4}$$

   with $|\psi_1\rangle = \begin{pmatrix} \cos\frac{\alpha}{2} \\ 0 \\ 0 \\ \sin\frac{\alpha}{2} \end{pmatrix}$ and $|\psi_2\rangle = \begin{pmatrix} -\sin\frac{\alpha}{2} \\ 0 \\ 0 \\ \cos\frac{\alpha}{2} \end{pmatrix}$. Note that members of this family include: $\rho(0) = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$ and $\rho(\pi/2) = \frac{1}{2}|\Phi^+\rangle\langle\Phi^+| + \frac{1}{2}|\Phi^-\rangle\langle\Phi^-|$, where $|\Phi^\pm\rangle = \frac{|00\rangle\pm|11\rangle}{\sqrt{2}}$ are entangled Bell states.

   (a) Considering the procedure for calculating operator expectation values in the density matrix formalism, discuss the use of measurements to observe the value of α.

   (b) Utilizing the perspective of Eq. (3) above, find the single-qubit measurement bases that connect $\rho(\alpha)$ to the state $|GHZ\rangle = \frac{|000\rangle+|111\rangle}{\sqrt{2}}$. In other words, in what basis could you measure a qubit in $|GHZ\rangle$ such that a combination of the measurement ensembles would produce the convex decomposition $\rho(0)$? $\rho(\pi/2)$? $\rho(\alpha)$?